

Multi-Platform Freelance Job Scraping Engine & Human-in-the-Loop Integration Architecture

A 16-platform data ingestion, anti-detection, normalization, and compliant delivery architecture for AI-assisted freelance proposal generation.

16

Platforms Integrated

5

Architecture Sections

4

Rollout Phases

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System Design & Compliance Architecture

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Classification

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- 1 16-Platform Data Ingestion Matrix & Analysis
- 2 Anti-Detection, Proxy & Scraping Infrastructure
- 3 Unified Data Normalization & Deduplication Engine
- 4 Terms of Service, Compliance & Account Safety
- 5 Implementation Roadmap & Technical Stack

EXECUTIVE SUMMARY

The system aggregates freelance job listings across 16 platforms of varying accessibility — from open REST/RSS feeds to session-locked, WAF-protected talent portals — normalizes them into a unified schema, deduplicates cross-posted briefs via semantic vector similarity, and delivers AI-drafted proposals through a browser extension that enforces human-in-the-loop submission to remain compliant with platform Terms of Service.

The architectural foundation of the multi-platform freelance job aggregator relies on a highly resilient, distributed data ingestion engine. Extracting structured job post data, client briefs, and project requirements across sixteen distinct platforms requires a highly heterogeneous approach. Platforms range from open, unauthenticated REST APIs to highly fortified, session-locked talent portals defended by enterprise-grade Web Application Firewalls (WAFs). The ingestion nodes must be decoupled from the core application logic, operating autonomously to fetch, normalize, and queue data for downstream AI processing.

1.1 Tier 1 — Open APIs & RSS Feeds

This category presents the lowest barrier to entry and the highest reliability. **Freelancer.com** operates a robust, publicly accessible REST API queryable via `GET` requests to `/api/projects/0.1/projects/active/`, supporting filters like `or_search_query` and returning structured JSON — eliminating fragile DOM parsing entirely. **Upwork** presents a dual-layered architecture: a strict OAuth2 GraphQL API for state-modifying actions, alongside public RSS feeds (e.g. `/ab/feed/jobs/rss`) that can be polled without triggering Cloudflare blocks. The GraphQL endpoint supports the `marketplaceJobPostings` query with the `USER_JOBS_SEARCH` type, given the correct read-scope permission. **PeoplePerHour** exposes listings via RSS or interception of internal AJAX/GraphQL calls. **Guru** exposes vertical-specific RSS feeds enabling lightweight polling. **Freelancermap** supports standard XML/Atom feeds, making extraction deterministic and layout-resistant.

1.2 Tier 2 — SPA DOM Scraping & XHR Interception

Behance loads project data via infinite scroll powered by internal XHR JSON APIs. **Contra**, a React/GraphQL SPA, requires headless browsers with network interception to capture and replay queries like `GetOpportunities`. **Fiverr** is protected by Datadome and PerimeterX, requiring full Playwright emulation; once JavaScript executes, the `__NEXT_DATA__` JSON object can be pulled directly from the page source. **99designs** lacks a public API and requires CSS-selector DOM parsing. **Legiit** and **Jobbers.io** use conventional marketplace structures parsable with basic HTTP clients, though headless rendering is recommended for lazy-loaded elements.

1.3 Tier 3 — Vetted, Session-Locked Talent Portals

Toptal is protected by aggressive Cloudflare/DataDome and behavioral analysis, requiring authenticated session injection into a stealth browser. **Codeable** requires persistent authentication with complex cookie/JWT management. **Lemon.io** and **Gun.io** similarly gate their project pipelines behind active developer profiles, requiring ingestion nodes to route through legitimately established sessions.

1.4 Tier 4 — Direct Channels & Social Streams

Reddit (`r/forhire`) is accessible via the official JSON API by appending `.json` to any subreddit URL, with minimal authentication for read access. **LinkedIn** public post streams, conversely, are defended by extreme

anti-bot ML models analyzing request cadence and network origin, typically requiring residential proxy networks and TLS fingerprint spoofing — or safer client-side browser-extension extraction.

1.5 Platform Extraction Summary

Platform	Extraction Path	Auth Level	Anti-Bot Defense	Discovery Latency	Fragility
Upwork	GraphQL API / RSS	OAuth2 / Public RSS	Cloudflare, TLS Fingerprinting	1–5 min	Medium
Freelancer.com	REST API	Public / Guest Token	Cloudflare, Rate Limiting	< 1 min	Low
Fiverr	Next.js JSON Extraction	Public / Session Cookies	Datadome, PerimeterX, CAPTCHA	5–15 min	High
PeoplePerHour	XHR Intercept / RSS	Public	Basic IP Rate Limiting	5 min	Low
Guru	RSS / DOM Parsing	Public	Akamai, Rate Limiting	5 min	Low
Contra	GraphQL Interception	Public	Cloudflare	10 min	Medium
Toptal	Authenticated DOM/ XHR	Vetted Session	Cloudflare, DataDome, Behavioral	15–30 min	Very High
Lemon.io	Authenticated XHR	Vetted Session	Custom Session Tracking	15–30 min	High
Gun.io	Authenticated XHR	Vetted Session	JWT Expiration, Rate Limiting	15–30 min	High
Codeable	Authenticated API	Vetted Session	Cloudflare, Strict Auth	15 min	High
99designs	DOM Parsing	Public	Fastly, Basic Fingerprinting	10 min	Medium
Behance	Internal XHR JSON	Public	Akamai	5 min	Low
Freelancermap	RSS / API	Public	Basic Rate Limiting	< 5 min	Low
Legiit	DOM Parsing	Public	Cloudflare (Standard)	10 min	Low
Jobbers.io	DOM Parsing	Public	Basic Rate Limiting	10 min	Low
Reddit	JSON API	Public / App Token	API Rate Limiting	< 1 min	Low
LinkedIn	HTML / XHR Intercept	Guest Tokens / Auth	Cloudflare, Datadome, ML Models	15–60 min	Extreme

1.6 Field Extraction Schema Strategy

Target Data Attribute	Extraction Strategy	Platform Example
Title & Description	DOM <code>div.job-title</code> , JSON keys, or markdown body parsing	Fiverr: <code>__NEXT_DATA__.props.pageProps.gig.title</code>
Budget / Hourly Rate	JSON extraction of nested currency & min/max constraints	Freelancer.com: <code>budget.minimum</code> , <code>budget.maximum</code>
Skills Required	Array mapping from embedded tags / GraphQL nodes	Upwork: <code>ontologySkills</code> array
Client Rating / Country	Extraction from nested poster metadata	Freelancer.com: <code>poster.country</code> , <code>poster.displayName</code>
Direct Job URL	String concatenation of base URI + canonical ID	Reddit: <code>reddit.com</code> + permalink
Posting Timestamp	ISO-8601 parsing for freshness / queue priority	Toptal/Freelancer: <code>postedAt</code> UNIX conversion

To sustain enterprise-grade ingestion across high-fragility platforms, the architecture layers proxy management, browser fingerprint evasion, and TLS-level spoofing. Standard HTTP clients face immediate termination from WAFs like DataDome and Akamai, which analyze cryptographic and behavioral fingerprints — necessitating a sophisticated spoofing framework.

2.1 Proxy Management Framework

Datacenter proxies handle unauthenticated, high-volume endpoints (Freelancer.com API, Upwork RSS, Reddit) where volumetric rate limiting dominates over deep behavioral analysis. **Residential proxies** are mandated for Fiverr, Contra, Behance, and LinkedIn, routing through legitimate ISPs with sticky sessions held for 10–30 minutes to avoid geolocation anomaly alerts. **Mobile proxies** (4G/5G, sitting behind Carrier-Grade NAT) are reserved for Tier 3 vetted portals — Toptal, Lemon.io, Codeable — since WAFs rarely ban mobile IPs shared by thousands of legitimate users.

2.2 Browser Fingerprint Evasion

Headless browsers (Playwright/Puppeteer) broadcast automation markers by default. `puppeteer-extra-plugin-stealth`-style middleware deletes `navigator.webdriver`, overrides `window.chrome`, and spoofs the Permissions API. WebGL/Canvas fingerprinting is defeated by injecting mathematically consistent noise into the rendering pipeline — unique per session, but internally stable within a session.

2.3 TLS / HTTP-2 Fingerprinting (JA3/JA4)

Security providers fingerprint the exact cipher sequence of the TLS handshake. Since Node.js/Python OpenSSL handshakes differ from Chrome/Safari's NSS/BoringSSL signatures, non-browser API requests are routed through TLS-spoofing clients such as `curl-impersonate` or `got-scraping`. The User-Agent, TLS fingerprint, Accept-Language headers, and HTTP/2 frame settings (TCP window size, pseudo-header order) must all stay internally consistent — any mismatch triggers immediate blocking.

2.4 Authenticated Session & Credential Vaulting

For Toptal/Codeable-class portals, login is never automated by scrapers. Credentials are resolved manually or via a secure Human-in-the-Loop relay; the resulting cookies/JWTs/LocalStorage state are encrypted in a Redis vault and injected into a fresh headless browser context via CDP on a high-trust mobile proxy at job time. Behavioral pacing — Bezier-curve mouse movement, 2,000–5,000 ms navigation pauses — simulates natural cognitive timing.

RISK NOTE

Vetted-portal sessions belong to real developer accounts; a ban is a severe operational failure. Mobile-proxy isolation and manual/HITL login resolution exist specifically to protect these accounts from automated-detection bans.

Disparate payload structures from sixteen platforms are transformed into a strictly typed, version-controlled schema before entering the AI proposal generation pipeline.

3.1 UnifiedJobPosting Schema

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "UnifiedJobPosting",
  "type": "object",
  "required": ["id", "platform", "source_url", "title",
    "description", "posted_at", "status"],
  "properties": {
    "id": { "type": "string", "description": "UUIDv4" },
    "platform": { "type": "string",
      "enum": ["upwork", "freelancer", "toptal", "codeable",
        "contra", "99designs", "reddit", "linkedin"] },
    "source_id": { "type": "string" },
    "source_url": { "type": "string", "format": "uri" },
    "title": { "type": "string" },
    "description": { "type": "string" },
    "budget": {
      "type": "object",
      "properties": {
        "type": { "type": "string",
          "enum": ["fixed", "hourly", "contest", "unspecified"] },
        "currency": { "type": "string", "maxLength": 3 },
        "min_amount": { "type": "number" },
        "max_amount": { "type": "number" }
      }
    },
    "skills": { "type": "array", "items": { "type": "string" } },
    "client": {
      "type": "object",
      "properties": {
        "client_id": { "type": "string" },
        "country": { "type": "string" },
        "rating": { "type": "number" },
        "history_spend": { "type": "number" }
      }
    },
    "posted_at": { "type": "string", "format": "date-time" },
    "ingested_at": { "type": "string", "format": "date-time" },
    "status": { "type": "string",
      "enum": ["active", "closed", "drafted"] }
  }
}
```

3.2 Cross-Platform Deduplication

Clients routinely cross-post identical briefs across platforms (e.g. Upwork and Freelancer.com) to maximize talent visibility. Hard heuristics first flag likely duplicates: identical title string, client country, and budget range

within a 48-hour window. Because titles and descriptions are often rephrased between platforms, a **semantic vector embedding pipeline** using a lightweight model (e.g. `all-MiniLM-L6-v2`) compares new job embeddings against all postings ingested in the trailing 72 hours. A **cosine similarity > 0.92** automatically links postings into a single `JobCluster` entity, so the dashboard surfaces each project once — letting the freelancer choose the platform with the most favorable commission.

3.3 Storage Architecture

Store	Role	Key Detail
PostgreSQL + pgvector	Primary operational database	JSONB payload with GIN indexing for schema-less nested queries; native <code><=></code> cosine similarity search
MongoDB	Cold-storage data lake	Archives raw, un-normalized HTML/JSON for replay if schema changes
Redis + BullMQ	In-memory pub/sub broker	Async queues decouple scraping nodes from normalization/embedding/AI-generation latency

4.1 Legal Frameworks

UNITED STATES — HIQ LABS V. LINKEDIN

Scraping publicly accessible data does not generally violate the CFAA, provided no authenticated barrier is circumvented without authorization.

INDIA — DPDPA 2023, SECTION 3(C)(II)

The Act does not apply to personal data made publicly available by the data principal — so extracting public client/freelancer data does not trigger DPDPA penalties.

EUROPEAN UNION — DSA / GDPR

Scraping for market research and lead generation remains viable under the DSA, but scraping private or restricted data without permission remains a severe GDPR violation.

Architectural directive: Tier 1–2 platforms are treated as public data repositories. Tier 3 vetted-portal extraction relies entirely on authorized user consent to access the user's own dashboard data.

4.2 The Air-Gap Principle

Server-side scraping infrastructure never uses the end-user's actual credentials. Public platforms are scraped unauthenticated or via synthetic burner accounts, keeping the backend ingestion engine fully isolated from user identity.

4.3 Human-in-the-Loop Delivery

AI-generated proposals are pushed via WebSocket to a Manifest V3 Chrome Extension running in the user's own authenticated browser session. Total automated bid submission violates virtually every platform's ToS and risks algorithmic bans, so the architecture strictly prohibits it. The extension only auto-fills the AI draft into the DOM for human review; **it must never programmatically click "Submit."** The human executes the final action — converting the workflow from prohibited bot automation into compliant AI-assisted drafting.

COMPLIANCE BOUNDARY

Automated pitch submission is strictly prohibited at the architectural level. The system is designed so that no code path can complete a bid without a human's manual click.

5.1 End-to-End System Architecture

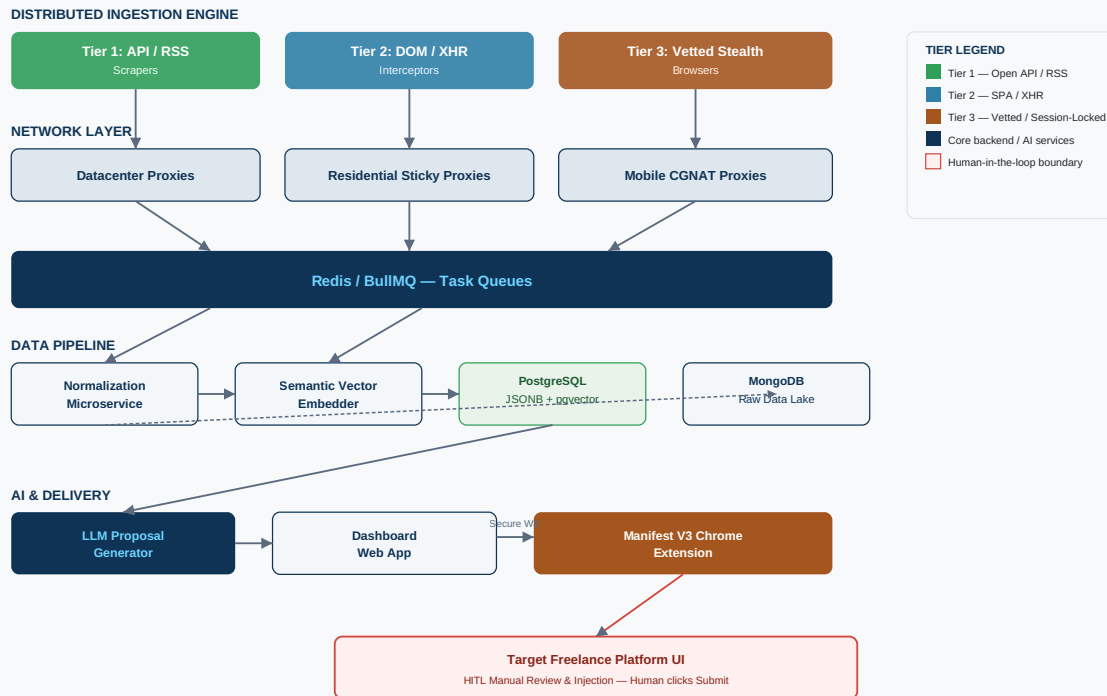


Figure 1 — Distributed ingestion nodes route through tier-matched proxies into a Redis/BullMQ queue, feeding the normalization & vector-embedding pipeline, PostgreSQL/pgvector store, and AI proposal generator, delivered to the user via a WebSocket-connected Manifest V3 extension for HITL review.

5.2 Technology Stack Matrix

Component	Technology	Justification
Backend API & Routing	Node.js (TypeScript)	High concurrency for non-blocking network I/O across thousands of simultaneous requests
DOM Scraping Engine	Playwright + Stealth Plugin	Full browser context emulation with CDP manipulation to bypass Datadome/PerimeterX challenges
HTTP Request Client	got-scraping / curl-impersonate	Precise JA3/JA4 TLS & HTTP/2 pseudo-header spoofing for non-headless API calls
Queue & Cache Broker	Redis + BullMQ	Absorbs ingestion bursts; coordinates distributed scraper nodes with guaranteed delivery
Primary Database	PostgreSQL + pgvector	JSONB flexibility plus native vector search — no separate vector DB needed

Component	Technology	Justification
Data Lake Archive	MongoDB	Scalable cold storage for raw HTML/JSON, enabling replay for schema fixes
AI Integration Layer	GPT-4o / Anthropic	Analyzes job constraints and drafts persuasive, context-aware proposals
Frontend UI	React / Next.js	Real-time WebSocket dashboard for deduplicated job streams
HITL Delivery Agent	Chrome Extension (Manifest V3)	Secure local DOM interfacing via service workers & content scripts

5.3 Phased Implementation Plan

PHASE 1 · WEEKS 1–4

Minimum Viable Product & Foundation (Tier 1)

Stand up PostgreSQL, MongoDB, and Redis; finalize the UnifiedJobPosting schema. Build scrapers for Upwork (RSS/API), Freelancer.com (REST API), PeoplePerHour, and Guru, using datacenter proxies. Deliverable: baseline AI-drafting pipeline and a rudimentary dashboard.

PHASE 2 · WEEKS 5–8

DOM Evasion & SPA Interception (Tier 2)

Deploy Playwright with residential sticky-session proxies. Reverse-engineer Next.js/React GraphQL states for Fiverr, Contra, Behance, Legiit, Jobbers.io, and 99designs, extracting from XHR intercepts. Introduce stealth-plugin modifications for mid-tier WAF bypass.

PHASE 3 · 5 WEEKS

High-Security Authentication & Vector Search (Tier 3–4)

Tackle Toptal, Codeable, Lemon.io, Gun.io, LinkedIn, and Reddit. Implement the encrypted Redis credential vault and mobile-proxy routing so vetted sessions are never tied to datacenter IPs. Finalize the `all-MiniLM-L6-v2` + pgvector semantic deduplication engine.

PHASE 4 · PRODUCTION SCALE

Human-in-the-Loop Integration & Production Scale

Finalize and ship the Manifest V3 Chrome Extension; establish secure WebSocket delivery of approved AI drafts. Write DOM-injection scripts tailored to each of the 16 platform UIs. Run

compliance audits confirming air-gap separation between scraping accounts and end-user identities before full production scale.